

Indian Institute of Technology Jodhpur
MAL1020
Tutorial Sheet 2

1. Prove that the quotient set V/W , the set of cosets $v + W$, is a vector space with following addition and scalar multiplication.

$$(v + W) + (u + W) = (u + v) + W, \quad a \cdot (v + W) = (a \cdot v) + W$$

2. Let $U = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1 + 2x_2 = 0 \text{ and } x_3 + 3x_4 = 0\}$. Find a basis and dimension of quotient space $\mathbb{R}^4/U$.
3. Suppose U and W are both 4-dimensional subspaces of $\mathbb{R}^6$. Prove that there exist two vectors in $U \cap W$ such that neither of these vectors is a scalar multiple of the other.
4. Let $U = \text{Span}\{(1, 1, 0, -1), (0, 1, 3, 1)\}$ and $W = \text{Span}\{(0, -1, -2, 1), (1, 2, 2, -2)\}$ be subspaces of $\mathbb{R}^4$. Find basis and dimension of $U + W$ and $U \cap W$.
5. Let P_n be the vector space of polynomials with real coefficients of degree less than or equal to n .
- (a). Let $U = \{p \in P_4 : p(2) = p(5) = p(6)\}$. Find a basis of U .
- (b). Extend the basis in part (a) to a basis of P_4 .
- (c). Find a subspace W of P_4 such that $P_4 = U \oplus W$.
6. Let V be a vector space and U_1, U_2, U_3 are subspaces. Prove or give a counterexample of the following:
- (a).

$$\text{Dim}(U_1 + U_2) = \text{Dim}(U_1) + \text{Dim}(U_2) - \text{Dim}(U_1 \cap U_2)$$

(b).

$$\begin{aligned} \text{Dim}(U_1 + U_2 + U_3) = & \text{Dim}(U_1) + \text{Dim}(U_2) + \text{Dim}(U_3) - \text{Dim}(U_1 \cap U_2) - \text{Dim}(U_2 \cap U_3) \\ & - \text{Dim}(U_1 \cap U_3) + \text{Dim}(U_1 \cap U_2 \cap U_3) \end{aligned}$$

7. Determine dimension of W , where $W = \{[x, y, z, w]^T \in \mathbb{R}^4 : w - z = y - x\}$
8. Find bases for the following subspaces of $\mathbb{R}^5$:

$$W_1 = \{(a_1, a_2, a_3, a_4, a_5) \in \mathbb{R}^5 : a_1 - a_3 - a_4 = 0\}$$

and

$$W_2 = \{(a_1, a_2, a_3, a_4, a_5) \in \mathbb{R}^5 : a_2 = a_3 = a_4 \text{ and } a_1 + a_5 = 0\}$$

What are the dimensions of W_1 and W_2 ?

9. Let

$$V = M_{2 \times 2}(\mathbb{R}), \quad W_1 = \left\{ \begin{pmatrix} a & b \\ c & a \end{pmatrix} \in V : a, b, c \in \mathbb{R} \right\}$$

and

$$W_2 = \left\{ \begin{pmatrix} 0 & a \\ -a & b \end{pmatrix} \in V : a, b \in \mathbb{R} \right\}.$$

Prove that W_1 and W_2 are subspaces of V , and find the dimensions of W_1 , W_2 , $W_1 + W_2$, and $W_1 \cap W_2$.

10. Let

$$V = M_{2 \times 2}(\mathbb{R}), \quad W_1 = \left\{ \begin{pmatrix} a & -a \\ c & d \end{pmatrix} \in V : a, b, c \in \mathbb{R} \right\}$$

and

$$W_2 = \left\{ \begin{pmatrix} a & b \\ -a & d \end{pmatrix} \in V : a, b \in \mathbb{R} \right\}.$$

Find basis and dimension of V/W_1 and V/W_2 .